

Research Methods — Module 03: Perception —

Study Questions

1. What neural signal corresponds to prediction errors in EEG? How is MMN generated?
2. How does MMN amplitude relate to prediction strength?
3. What oscillatory band reflects precision modulation? How does alpha power encode precision?
4. How does the roving oddball paradigm test Active Inference's account of belief updating?
5. What is the directional flow hypothesis? Which frequencies carry bottom-up vs. top-down signals?
6. Design a hierarchical prediction paradigm. What PE signatures would you expect at each level?
7. How can laminar recordings distinguish prediction errors from predictions?
8. What are error neurons, prediction neurons, and precision neurons?
9. What does Granger causality measure? How is it applied to frequency-resolved signals?
10. Describe DCM for EEG. How does it differ from DCM for fMRI?
11. What is a neural mass model? How does it serve as the generative model in DCM for EEG?
12. How does time-frequency analysis reveal the temporal dynamics of predictive processing?
13. What does source localization add beyond scalp topography?
14. How does volatility manipulation affect prediction error responses?
15. How does attention modulate oscillatory signatures in an Active Inference framework?
16. What is steady-state response (SSR)? How can it be used to probe predictions?
17. How do interhemispheric differences in alpha reflect lateralized precision allocation?
18. Compare the temporal resolution of EEG vs. fMRI for testing Active Inference predictions.
19. What methodological artifacts can contaminate prediction error measurements?
20. Design an electrophysiology experiment testing a specific Active Inference prediction about perception.